Immanuel Peter

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EDUCATION

The University of Chicago – **BS Computer Science, BA Physics** | *expected June 2028*

EXPERIENCE

Member of Technical Staff Intern | Tensormesh | *March 2026 – Present*

- Integrated Arize Phoenix and emitted OpenInference traces for all router requests, with SDK and CLI tools for trace/span inspection.
- Built an orchestrator that asynchronously pins prompts in LMCache during inference, exposed through the API, SDK, and CLI.
- Added upstream LMCache support for listing pinned entries and tracking per-key access in the coordinator.
- Led an internal research effort on key weighting for long-context inference.

Software Engineer Intern | Quantum Rings | *June 2025 – August 2025*

- Reduced API request latency by designing queue-driven telemetry processing that decoupled heavy operations from the API, enabling horizontal scaling (AWS SQS, TypeORM)
- Built full-stack admin analytics dashboards tracking user growth and execution volume with SQL time-bucket aggregation and time zone safe filtering (Next.js, NestJS, Recharts)

OPEN SOURCE

LMCache - Python | github.com/LMCache/LMCache

- Added a coordinator endpoint to list pinned cache entries. #4960
- Added per-key access tracking to the coordinator key directory. #4873

Brev CLI – Go | github.com/brevdev/brev-cli

- Authored rsync-first transfers with SCP fallback. #423

vLLM Production Stack - Kubernetes | http://github.com/vllm-project/production-stack

- Fixed default runtimeClass to nvidia in VllmRuntime operator; regenerated CRD manifests. #974
- Added extraVolumes and extraVolumeMounts to routerSpec in Helm chart for custom mounts. #975

PROJECTS

Vision Tower Bench | ipeter.dev/blog/vision-tower-bench

- Built a benchmark for eleven vision towers across recognition, geometry, correspondence, and occlusion, with paired bootstrap intervals.
- Extracted and published nine standalone vision towers (Gemma 4, Qwen3.8, DeepSeek V4.1, Nemotron Omni, and more) on Hugging Face, verified to BF16 parity with their parent checkpoints.
- Built the adapter, feature-cache, and probe pipeline with capacity-matched PCA heads, per-cell LR selection, and 3-seed runs across 8 tapped depths.

AutoMoE — MoE Self-Driving Model | github.com/immanuel-peter/self-driving-model

- Built a 53M parameter modular self-driving stack in PyTorch with 4 expert networks, top-2 gating, and 10-waypoint trajectory prediction.
- Released ~145K CARLA frames across multimodal and multi-camera datasets, and trained/fine-tuned the pipeline with PyTorch DDP on up to 8 GPUs.

CARLA Autopilot Datasets | hf.co/immanuelpeter/datasets | 19K+ Downloads

- Released two MIT-licensed CARLA autopilot datasets (~145K frames, 567 GB) on Hugging Face with 2.2K and 6.5K downloads each; flagship split includes synchronized 4-camera RGB, 32-channel LiDAR, semantic segmentation, and 2D bounding boxes across 24 weather/map scenarios
- Designed run-based train/val/test splits to eliminate temporal leakage, supporting perception, sensor fusion, imitation learning, and RL benchmarking under extreme conditions

Hostess — Docker Compose for Production | hostess.sh

- Designed and built a multi-cloud PaaS control plane (Go CLI/API + Kubernetes controller + Next.js Studio) that compiles a declarative config into production workloads
- Implemented the deploy pipeline end-to-end: multi-backend image builds, dependency-ordered manifest generation, secret injection, readiness waits, and stable URL routing